
Tools/Modules

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Introduction

Tools installed on the NGC-HPC or an NGC-hosted Self-manged cloud are managed using modules. At the moment NGC offer around 4200 installed modules, also counting submodules and packages.

The modules contain:

Bioinformatic tools like: Samtools, BWA, GATK, VCFtools, AdapterRemoval, ANNOVAR

Development tools like: Visual Studio Code, Vim, Emacs, Perl, Python, R, Nextflow, Snakemake, Anaconda

And more!

See below for useful commands in relation to listing available modules and loading/unloading modules into your environment.

Why modules

Modules provide a mechanism to set/unset all environment variables related to a given package in one UNIX shell/terminal command. Module definitions are contained in so-called

module files located in several directories. The directory structure is dependent on the system.

User creation and modification

To access modules you need to become a diagnostic user (clinical services) or research user (NGC-managed cloud or Self-managed cloud) of the NGC services.

[Here](#) (link only accessible from the NGC Wiki) you can read about how to become a user of NGC services.

As a user on the NGC-HPC or an NGC-hosted self-managed cloud you have access to all available modules by default.

Logging into the system for the first time

No special requirements.

Username and password

There is no password restriction on this solution as long as you are logged in to Citrix-NGC (<https://secureremote.ngc.dk>) and NGC-HPC or VMWare Horizon and a Self-manged cloud, you can access the modules (see details below).

How to use the system

1. Connect to a login node

From Citrix-NGC

This solution is for diagnostic (clinical services) users and research users on **NGC-managed cloud**.

First you need to log in to Citrix-NGC (<https://secureremote.ngc.dk>). [Here](#) (link only accessible from the NGC Wiki) is a guide on how to log in to Citrix-NGC.

Then you need to access NGC-HPC.

Alternative 1) Open the MobaXterm app and log in to NGC-HPC through one of the login nodes.

Alternative 2) Open NGC Desktop, then open MobaXterm from the desktop icon and log in to NGC-HPC through one of the login nodes.

See this guide for more details on how to complete the above steps: [NGC Desktop and NGC-HPC](#) (link only accessible from the NGC Wiki)

From Self-manged cloud

This solution is for research users with access to **Self-managed cloud**.

Open VMware Horizon Client (<https://spc.ngc.dk> is the connection server).

Enter your NGC username and password and click **Login**.

You will receive a notification on your phone from Entrust Identity app. Click on the app notification, login with your chosen PIN (or face recognition) and choose **Confirm**.

You are now logged into VMware Horizon client and can see an overview of the services you have access to.

Double click on COS-CLDXXX to login to a CentOS desktop.

Open the Terminal (double click the Terminal icon on the desktop) and ssh into a login node. Enter your password.

```
# ssh into a Login node
ssh -X <username>@cld<cloud_id>-0001
# example with user jondoe on cCloud 099
ssh -X jondoe@cld099-0001
```

The **-X** option makes it possible to open graphical user interface tools like e.g. Firefox from your session.

2. Start using the modules

Logged into a login node on NGC-HPC or NGC-hosted self-managed cloud, you have access to the modules.

Useful commands for modules

In the terminal you can write the following commands to list, display, load etc. available modules.

Lines starting with # are comments, which are ignored by the bash shell (they are not executed commands).

Module commands

```
# get a list of all the available modules on the cloud/NGC-HPC
module avail
# press "down arrow" to scroll and "q" to exit

# searching for a specific module
module avail <module name>
# e.g.
module avail bwa
# note: the shell/terminal is case sensitive why "module avail BWA" will not result
in any hits

# get basic information on module files
module whatis <module name/version>
# e.g.
module whatis bwa/0.7.17
# note: for several of the commands you need to include the version of the tool
(see available versions using "module avail <module name>")

# Load a module to get access to the tool and its functionality
module load <module name/version>
# e.g.
module load bwa/0.7.17

# info and options for the BWA tool
man bwa

# for many tools you can also get help by including the option -h or --help after
having loading the tool
# e.g.
module load anaconda3/2022.10
anaconda --help

# get further information with "module display"
```

```
# "module display" will list conflict - module(s) which conflict with this module
and can't be loaded at the same time
# "module display" will list prereq - modules which must be loaded to use this
module
module display <module name/version>
# e.g.
module display vcftools/0.1.16

# to load vcftools you need to first load "perl", which in its turn has the prereq
"tools"
# therefore to load vcftools:
module load tools
module load perl/5.30.2
module load vcftools/0.1.16
# you can also load all necessary modules in one line
module load tools perl/5.30.2 vcftools/0.1.16
# note: if there is more than one tool written on the same line after prereq, you
only need to load one of them

# search for a particular string in the basic information of the module files
module apropos <string>
# e.g.
module apropos conda

# List Anaconda modules (i.e., Python modules)
module load anaconda2/4.4.0
conda list

# List Perl modules - Listing software not included as modules
module load tools perl/5.20.1
module display perl/<version>

# to see available packages in R
module display R/<version>
# e.g.
module display R/4.2.0
# from here you can see location of libraries (packages)
ls /services/tools/R/4.2.0/lib64/R/library
# no. of Libraries
ls /services/tools/R/4.2.0/lib64/R/library | wc -l
# search for a specific library
ls /services/tools/R/4.2.0/lib64/R/library | grep ggplot2

# see which modules you have loaded into the environment
module list

# create a saved default environment - load all your desired modules and then type
```

```
module save
```

```
# to save a set of non-default modules, you need to assign a environment name
```

```
module save <environment-name>
```

```
# Load all modules of a default saved environment
```

```
module restore
```

```
# Load modules of a specific saved environment
```

```
module restore <environment-name>
```

```
# get a List of your saved environments
```

```
module savelist
```

```
# how to unload a module?
```

```
module unload <module name/version>
```

```
# swap a specific module for another one (especially useful to switch between different versions of the same program)
```

```
module switch <old_module> <new_module/version>
```

```
# e.g.
```

```
module switch bamtools samtools/1.14
```

```
# unload all modules
```

```
module purge
```

```
# manage modules in your login environment (OBS - keep it to an absolute minimum)
```

```
# add modulefile(s) to the shell's initialization file in the user's home directory
```

```
initadd <module name>
```

```
# does the same as initadd but prepends the given modules to the beginning of the list
```

```
initprepend modulefile <module name>
```

```
# remove modulefile(s) from the shell's initialization files
```

```
initrm <module name>
```

```
# switch modulefile1 with modulefile2 in the shell's initialization files
```

```
initswitch <module name 1> <module name 2>
```

```
# List all of the modulefiles loaded from the shell's initialization file
```

```
initlist
```

```
# clear all of the modulefiles from the shell's initialization files
```

```
initclear
```

```
# get more module options
```

```
man module
```

Note

It is not recommended to run compute heavy commands on the login node. You can submit a command for running a tool as a job using qsub. See [Submitting jobs to the queue system](#) (link only accessible from the NGC Wiki)

Conclusion

You should now be able to see available modules and load a module/tool for usage.

Relevant links

How to submit a job to the queue system: [Submitting jobs to the queue system](#) (link only accessible from the NGC Wiki)

Error handling

Nothing relevant.

Support

You can get help with issues related to the NGC services by writing an e-mail to support@ngc.dk. You can significantly speed up the support process by providing complete and detailed information for faster troubleshooting and error correction.

In the email to support@ngc.dk please include the following information (as detailed as you can in relation to your knowledge):

- An informative e-mail subject line.
- Your name, NGC username, and the [project \(user group\)](#) you belong to.
- When did the problem occur, how often does it occur and duration of issue.
- Which cluster? Which node? Which job?
- What happened? What exactly were you doing/trying to do? Include information on

which commands you were using and/or which tools/modules (incl. versions).